

# Transformations, Compositions and Inverses Together

---

*Transforming a rule, composing two functions, and inverting a function*

**Before you start.** All three ideas on this sheet are the same idea seen from different sides: a transformation composes  $f$  with a simple linear map, and an inverse undoes a composition.

- **Inside the function** changes  $x$  and acts *horizontally*, in the direction opposite to the sign:  $f(x - h)$  shifts right by  $h$ ,  $f(kx)$  compresses by a factor  $1/k$ .
- **Outside the function** changes  $y$  and acts *vertically*, exactly as written:  $af(x) + c$  stretches by  $a$  and shifts up by  $c$ .
- **Composition works inside out:**  $(f \circ g)(x) = f(g(x))$ , so evaluate  $g$  first.
- **An inverse undoes:**  $f^{-1}(f(x)) = x$  and  $f(f^{-1}(x)) = x$ . To find it, swap  $x$  and  $y$  and solve for  $y$ .

**1.** Let  $f(x) = x^2$ . Write the rule for  $g(x) = f(x - 2) + 5$  as a single expanded polynomial, and say which direction each shift goes.

Answer: \_\_\_\_\_

**2.** Let  $f(x) = 2x - 3$  and  $g(x) = x^2 + 1$ . Find  $(f \circ g)(2)$ . Show the inner value before you use it.

Answer: \_\_\_\_\_

**3.** Let  $f(x) = 2x - 3$ .

(a) Find  $f^{-1}(x)$  by swapping  $x$  and  $y$  and solving.

(b) Check your answer by simplifying  $f(f^{-1}(x))$ , and evaluate  $f^{-1}(7)$ .

Answer: \_\_\_\_\_

**4.** Let  $f(x) = x^2 - x$ . Write the rule for  $h(x) = -f(2x)$  as an expanded polynomial, and name the two transformations involved.

Answer: \_\_\_\_\_

**5.** Let  $f(x) = x + 4$  and  $g(x) = x^2$ . Write  $(g \circ f)(x)$  as an expanded polynomial. Then say how the graph of  $g \circ f$  is related to the graph of  $g$ .

Answer: \_\_\_\_\_

**6.** Let  $f(x) = \frac{2x+1}{x-3}$ .

(a) Find  $f^{-1}(x)$ . (Swap  $x$  and  $y$ , clear the denominator, then collect the  $y$  terms on one side.)

(b) Verify by simplifying  $f(f^{-1}(x))$ , and evaluate  $f^{-1}(4)$ .

Answer: \_\_\_\_\_

**7.** The graph of  $h$  is the graph of  $f(x) = \sqrt{x}$  stretched vertically by a factor of 3, then shifted left 4 and down 1.

(a) Write the rule for  $h(x)$ .

(b) Find  $h(5)$ .

Answer: \_\_\_\_\_

**8.** Let  $f(x) = \frac{1}{x-2}$  and  $g(x) = x + 5$ . Find  $(f \circ g)(-1)$ , and state the one input value that  $f \circ g$  must exclude from its domain.

Answer: \_\_\_\_\_

**9.** Let  $f(x) = 3x - 7$ . For which value of  $x$  is  $f^{-1}(x) = 2$ ? Answer without ever writing  $f^{-1}$  if you can see the shortcut, then check with the formula.

Answer: \_\_\_\_\_

**10.** Let  $f(x) = 2x$  and  $g(x) = x + 3$ .

(a) Write  $(f \circ g)(x)$  and then find its inverse.

(b) Separately, compute  $g^{-1}(f^{-1}(x))$ . Show that the two results agree, and say in one sentence why the order of  $f$  and  $g$  reverses.

Answer: \_\_\_\_\_

- 11.** A table gives two values of  $f$ :  $f(1) = 4$  and  $f(3) = -2$ . Let  $h(x) = -2f(x - 1) + 3$ .
- (a) Find  $h(2)$ .
- (b) Describe, in order, the three transformations that carry the graph of  $f$  to the graph of  $h$ .

Answer: \_\_\_\_\_

**12. Synthesis.** Let  $f(x) = (x - 1)^2$  with the domain restricted to  $x \geq 1$ .

- (a) Explain why this restriction gives  $f$  an inverse, when  $f$  on all real numbers has none.
- (b) Find  $f^{-1}(x)$  and verify it by simplifying  $f(f^{-1}(x))$ .
- (c) Compute  $f^{-1}(f(5))$ , and say what that value shows about reflecting the graph across the line  $y = x$ .

Answer: \_\_\_\_\_